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Mud guard control systems for vehicles and methods of controlling mud guards

Abstract

A vehicle includes a mud guard control system including a speed sensor configured to provide a signal indicative of speed to a controller. The controller is configured to instruct an actuator to move a mud guard from a stowed configuration to a deployed configuration based on the signal from the speed sensor when the speed of the vehicle is below a predetermined threshold value.

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Background/Summary

TECHNICAL FIELD

(1) The present specification generally relates to mud guard control systems for vehicles and, in particular mud guard control systems that extend and retract mud guards for vehicles.

BACKGROUND

(2) Mud guards are often flaps that are suspended behind the rear wheels of vehicles. Mud guards are commonly used as they can protect pedestrians, adjacent vehicles and the vehicles themselves from mud slung from the wheels. Because mud guards are typically flat, solid pieces of material that are suspended vertically, the mud guards can increase drag on the vehicles, which can reduce fuel efficiency. The fuel efficiency can be particularly affected by the presence of the mud guards when the vehicles reach higher speeds.

(3) Accordingly, a need exists to provide mud guard control systems that control operation of mud guards.

SUMMARY

(4) In one embodiment, a vehicle includes a mud guard control system including a speed sensor configured to provide a signal indicative of speed to a controller. The controller is configured to instruct an actuator to move a mud guard from a stowed configuration to a deployed configuration based on the signal from the speed sensor when the speed of the vehicle is below a predetermined threshold value.

(5) In another embodiment, a method of controlling operation of a mud guard using a mud guard control system of a vehicle is provided. The method includes a controller receiving a signal indicative of speed of the vehicle from a speed sensor. If the speed of the vehicle is below a predetermined threshold value determined using the signal, the controller instructs an actuator to move the mud guard from a stowed configuration to a deployed configuration.

(6) These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

(2) FIG. 1 depicts a vehicle including a mud guard control system for controlling operation of mud guards, according to one or more embodiments shown and described herein;

(3) FIG. 2 depicts the mud guard control system of FIG. 1, according to one or more embodiments shown and described herein; and

(4) FIG. 3 depicts a method of controlling a mud guard using a mud guard control system, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

(5) Embodiments described herein are generally related to vehicles that include mud guard control systems. The mud guard control systems include a controller that controls operation of an actuator so as to extend a mud guard when a vehicle is operating below a preselected speed and to retract the mud guard when the vehicle is operating above the preselected speed. The mud guard control systems may also control operation of a guard engagement device adjacent the mud guard that removes debris from the mud guard.

(6) Referring to FIG. 1, a portion of a vehicle **10** is illustrated that includes a mud guard control system **12** for use in controlling mud guards **14** of the vehicle **10**. While a bed portion **16** of a truck is shown, it is noted that the present disclosure may be applied to any vehicle, such as semi-trailer trucks, all-terrain vehicles, utility task vehicles and cars. It is contemplated that the vehicle may or may not be an autonomous vehicle or a partially autonomous vehicle. The vehicle may be fully drivable.

(7) The vehicle **10** includes a plurality of wheel wells including rear wheel wells **16**. The wheel wells **16** are locations where wheels **18** of the vehicle **10** are located. The mud guard control system **12** includes the mud guard **14**, an actuator **22** that is operatively connected to the mud guard **14** for raising and extending the mud guard **14** and a guard engagement device **24** (e.g., a bar or squeegee) that can be placed in engagement with the mud guard **14** for removing mud and debris from the mud guard **14**. In some embodiments, the actuator **22** may move the guard engagement device **24** between a retracted position **24a** and an engaged position **24b** in engagement with the mud guard **14**. When the mud guard **14** is raised using the actuator **22**, the mud guard **14** can scrape against the guard engagement device **24**, which removes some of the mud and debris from the mud guard **14**. The mud guard **14** can also be raised into a housing **25** that is located within a body **27** of the vehicle **10**.

(8) Referring to FIG. 2, the vehicle **10** includes the mud guard control system **12** that includes one or more processors **40**. Each one or more processor **40** may be any device capable of executing machine readable instructions. Accordingly, each one or more processor **40** may be a controller, an integrated circuit, a microchip, a computer, or any other computing device, collectively referred to as controllers. The controllers **40** are coupled to a communication path **52** that provides signal interconnectivity between various modules. Accordingly, the communication path **52** may communicatively couple any number of controllers **40** with one another, and allow the modules coupled to the communication path **52** to operate in a distributed computing environment. Specifically, each of the modules may operate as a node that may send and/or receive data. As used herein, the term “communicatively coupled” means that coupled components are capable of exchanging data signals with one another such as, for example, electrical signals via conductive medium, electromagnetic signals via air, optical signals via optical waveguides, and the like.

(9) Accordingly, the communication path **52** may be formed from any medium that is capable of transmitting a signal such as, for example, conductive wires, conductive traces, optical waveguides, or the like. In some embodiments, the communication path **52** may facilitate the transmission of wireless signals, such as WiFi, Bluetooth, and the like. Moreover, the communication path **52** may be formed from a combination of mediums capable of transmitting signals. In one embodiment, the communication path **52** comprises a combination of conductive traces, conductive wires, connectors, and buses that cooperate to permit the transmission of electrical data signals to components such as processors, memories, sensors, input devices, output devices, and communication devices. Accordingly, the communication path **52** may comprise a vehicle bus, such as for example a LIN bus, a CAN bus, a VAN bus, and the like. Additionally, it is noted that the term “signal” means a waveform (e.g., electrical, optical, magnetic, mechanical or electromagnetic), such as DC, AC, sinusoidal-wave, triangular-wave, square-wave, vibration, and the like, capable of traveling through a medium.

(10) The vehicle **10** further includes one or more memory modules **44** coupled to the communication path **52**. The one or more memory modules **44** may comprise RAM, ROM, flash memories, hard drives, or any device capable of storing machine readable instructions such that the machine readable instructions can be accessed by the one or more processors **40**. The machine readable instructions may comprise logic or algorithm(s) written in any programming language of any generation (e.g., 1GL, 2GL, 3GL, 4GL, or 5GL) such as, for example, machine language that may be directly executed by the processor, or assembly language, object-oriented programming (OOP), scripting languages, microcode, etc., that may be compiled or assembled into machine readable instructions and stored on the one or more memory modules **44**. Alternatively, the machine readable instructions may be written in a hardware description language (HDL), such as logic implemented via either a field-programmable gate array (FPGA) configuration or an application-specific integrated circuit (ASIC), or their equivalents. Accordingly, the methods described herein may be implemented in any suitable computer programming language, as pre-programmed hardware elements, or as a combination of hardware and software components.

(11) The vehicle **10** comprises a display **48** for providing visual output such as, for example, maps, navigation, entertainment, information, or a combination thereof. The display **48** is coupled to the communication path **52**, as shown in FIG. 2. Accordingly, the communication path **52** communicatively couples the display **48** to other modules of the vehicle **10**. The display **48** may include any medium capable of transmitting an optical output such as, for example, a cathode ray tube, light emitting diodes, a liquid crystal display, a plasma display, or the like. Moreover, the display **48** may be a touch screen that, in addition to providing optical information, detects the presence and location of a tactile input upon a surface of or adjacent to the display **48**. Accordingly, the display **48** may receive mechanical input directly upon the optical output provided by the display **48**. The display **48** can be used to control operation of the mud guard control system **12**. It is noted that the display **48** can include at least one of the one or more processors **40** and the one or memory modules **44**. The display **48** can be at least one of a heads-up display, an instrument cluster display, and a mobile device display. In some embodiments, the vehicle **10** may have a plurality of displays. In such embodiments, the vehicle **10** can also have a plurality of different types of displays at various locations within the vehicle **10**. For example, and not as a limitation, the vehicle **10** can have an in-dashboard display and a heads-up display for displaying information directly on a windshield or window of the vehicle **10**.

(12) In some embodiments, the vehicle **10** comprises network interface module **50** for communicatively coupling the vehicle **10** to a network **55**. The network interface module **50** can be communicatively coupled to the communication path **52** and can be any device **54** capable of transmitting and/or receiving data via the network. Accordingly, the network interface module **50** can include a communication transceiver for sending and/or receiving any wired or wireless communication. For example, the network interface module **50** may include an antenna, a modem, LAN port, Wi-Fi card, WiMax card, mobile communications hardware, near-field communication hardware, satellite communication hardware and/or any wired or wireless hardware for communicating with other networks and/or devices. In one embodiment, the network interface module **50** includes hardware configured to operate in accordance with the Bluetooth wireless communication protocol. In another embodiment, network interface module **50** may include a Bluetooth send/receive module for sending and receiving Bluetooth communications to/from a mobile device **54** (e.g., a smartphone including a virtual key) for controlling operation of the mud guard control system **12**.

(13) The vehicle **10** may include one or more sensors, generally referenced as element **56**, communicatively coupled to the one or more processors **40**. The one or more sensors **56** may include, but are not limited to, speed sensors, cameras, LiDAR, RADAR, proximity sensors, pressure sensors, etc. In some embodiments, multiple types of sensors are used to provide a variety of information to the vehicle **10**, such as vehicle speed.

(14) One or more actuators **22** and **60** are communicatively coupled to the communication path **52** so as to be controlled by the controller **40**. The actuator **22** may be operatively connected to the mud guard **14** for raising and lowering the mud guard **14** and the actuator **60** may be operatively connected to the guard engagement device **24** for moving the guard engagement device **24** into and out of engagement with the mud guard **14**. Any suitable actuators **22** and **60** can be used, such as motors, pneumatic and/or hydraulic cylinders, etc. For example, the actuator **22** may include a take up wheel **62** (FIG. 1) that can be used to roll up the mud guard **14** when not in use.

(15) Referring to FIG. 3, a method **100** of controlling operation of the mud guard **14** using the mud guard control system **12** is illustrated. At step **102**, the controller **40** receives signals indicative of speed of the vehicle from the speed sensor **56**. If the speed of the vehicle is below a predetermined threshold value (e.g., 35 mph or less, such as 30 mph or less, such as 25 mph or less, such as 20 mph or less, such as 15 mph or less, such as 10 mph or less), the controller **40** instructs the actuator **22** to lower the mud guard **14** to the deployed configuration automatically using logic saved in the memory **44** at step **104**. At step **106**, the controller **40** continues to monitor the signals from the

speed sensor **56**. If the speed of the vehicle is above the predetermined threshold value, the controller instructs the actuator **22** to raise the mud guard **14** to the stowed configuration at step **108**. As or before the mud guard **14** is raised, the actuator **60** may move the guard engagement device **24** into engagement with the mud guard **14** at step **110**. Movement of the mud guard **14** against the guard engagement device **24** may cause mud and debris on the mud guard **14** to scrape against the guard engagement device **24** thereby removing mud and debris therefrom.

(16) The above-described mud guard control systems can be used to automatically extend and retract mud guards between deployed and stowed configurations. The mud guards can be deployed and retracted automatically based on vehicle speed, which can allow the mud guard to be deployed at relatively low vehicle speeds and retracted at relatively high vehicle speeds. This control arrangement can reduce the effect on drag that the mud guard has at the relatively higher vehicle speeds. The mud guard has less influence on coefficient of drag at lower vehicle speeds when it is in the deployed configuration.

(17) While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A vehicle comprising: a mud guard control system comprising a speed sensor configured to provide a signal indicative of speed of the vehicle to a controller; wherein the controller is configured to instruct an actuator to move a mud guard from a stowed configuration to a deployed configuration based on the signal from the speed sensor when the signal provided by the sensor indicates to the controller that the speed of the vehicle is below a predetermined threshold value; and a guard engagement device that is configured to remove mud from the mud guard when the mud guard is moved from the deployed configuration to the stowed configuration, the guard engagement device configured to move from a retracted position removed from the mud guard to an engaged position in engagement with a tire facing side of the mud guard based on instructions from the controller.
2. The vehicle of claim 1 further comprising another actuator configured to move the guard engagement device from the retracted position removed from the mud guard to the engaged position in engagement with the mud guard.
3. The vehicle of claim 1 further comprising a housing in a body of the vehicle configured to receive the mud guard in the stowed configuration.
4. The vehicle of claim 1, wherein the controller is configured to instruct the actuator to move the mud guard from the deployed configuration to the stowed configuration based on the signal from the speed sensor when the speed of the vehicle is above the predetermined threshold value.
5. A method of controlling operation of a mud guard using a mud guard control system of a vehicle, the method comprising: a controller receiving a signal indicative of speed of the vehicle from a speed sensor; when the speed of the vehicle is below a predetermined threshold value determined by the controller using the signal from the speed sensor, the controller instructing an actuator to move the mud guard from a stowed configuration to a deployed configuration; moving a guard engagement device from a retracted position removed from the mud guard to an engaged position in engagement with a tire facing side of the mud guard based on instructions from the controller; and removing mud from the mud guard using the guard engagement device when the mud guard is moved from the deployed configuration to the stowed configuration.
6. The method of claim 5 further comprising another actuator moving the guard engagement device

from the retracted position removed from the mud guard to the engaged position in engagement with the mud guard.

7. The method of claim 5 further comprising a housing in a body of the vehicle configured to receive the mud guard in the stowed configuration.

8. The method of claim 5, wherein the controller instructing the actuator to move the mud guard from the deployed configuration to the stowed configuration based on the signal from the speed sensor when the speed of the vehicle is above the predetermined threshold value.

9. The method of claim 8, wherein the mud guard is moved into a body of the vehicle in the stowed configuration.
